Battery literature digest — week of 30 October 2026

9 papers selected from 1251 scanned. Scores out of 10; click a title to open the paper.

1. Resolving the Structural Origin of Lattice Oxygen Release across Scales in Practical Lithium-Manganese-Rich Electrodes

8/10 · *Advanced Energy Materials* · OEMS + multiscale synchrotron X-ray spectro-microscopy (XANES mapping) + STEM/EELS

Maps Mn reduction as a fingerprint of O2 release from electrode to grain scale (four orders of magnitude). It finds no electrode- or particle-scale gradient. Frontal (Li-intercalation-plane) facets reduce more deeply, and normalizing by frontal-facet area reconciles the twofold difference in O2 evolution between the two electrodes. This gives a quantitative structural descriptor.

2. Discovering Underpotentially Deposited Crystalline Sodium Metal Nanoparticles on the Surface of Hard Carbon Anodes

8/10 · *Journal of the American Chemical Society ASAP* · cryo-EM + cryo-electron tomography (EELS/diffraction)

Cryo-ET localizes crystalline Na metal clusters in sodiated hard carbon in 3D, finding them on or near particle surfaces rather than in closed bulk nanopores. This contradicts the prevailing pore-filling picture and redirects mitigation of underpotential plating toward particle surfaces and interfaces.

3. Monitoring Lithiation Gradients in Solid-State Batteries Using Operando Neutron Radiography

7/10 · *Energy Storage Materials* · operando neutron radiography

Operando neutron radiography is used to map lithiation gradients across solid-state battery electrodes, exploiting the strong neutron contrast of Li. This is a relevant, uncommon operando imaging method for SSBs. The abstract is missing, so quantitative findings can’t be judged.

4. Operando X-ray Photoelectron Spectroscopy Reveals Oxide Solid Electrolyte Stability against Electrochemically Plated Lithium

7/10 · *ACS Energy Letters ASAP* · lab-based operando XPS with electrochemical Li plating

Plates Li electrochemically inside the XPS, avoiding the artefacts of evaporation or sputtering. It shows LATP is reduced to a Li+/Ti3+ mixed-conducting interphase that partly relaxes, while LLZO shows no resolvable Zr or La reduction. This resolves a known inconsistency in the in situ XPS literature.

5. Operando Magnetometry Decodes Space-Charge Storage in the Solid-Electrolyte Interphase

7/10 · *Advanced Materials* · operando magnetometry, operando AP-XPS

Uses operando magnetometry with Fe3C as a magnetic probe to follow reversible charge storage at the SEI/electrode interface. Operando AP-XPS supports an SEI-centred space-charge mechanism, giving about 236 mAh/g extra capacity. The method is new and transferable, but space-charge storage claims in conversion-type electrodes are still debated and the evidence is indirect.

6. Demonstrating the Generality of Kerr-Gated Raman Spectroscopy Measurements of High Charge States in Graphitic Electrodes for Li-Ion Batteries

6/10 · *ACS Applied Energy Materials ASAP* · operando Kerr-gated Raman spectroscopy

Kerr-gating suppresses fluorescence and lets the graphite G-band be resolved at high lithiation (stage 2→1, $0.5 \leq x \leq 1$), where conventional Raman fails. The paper shows the G-band shift tracks state of lithiation across graphite sources, carbonate and ionic-liquid electrolytes, and 10 cycles. The work is a generality demonstration, not spatial imaging.

7. Operando Raman Spectroscopy-Guided 24-Hour Aging Assessment of Activated Carbon Positive Electrodes for Lithium-Ion Capacitors

6/10 · *ACS Energy Letters ASAP* · operando Raman spectroscopy (D1 band analysis)

Operando Raman of AC cathodes in lithium-ion capacitors under high-voltage cycling finds a turning point in D1 band width. It narrows first as CEI and gas passivate defects, then broadens with corrosion. An earlier turning point correlates with faster capacity decay, giving a 24-h aging diagnostic. The work is spectroscopic but not spatially resolved.

8. In Situ Visualizing the Electric-Field-Driven Assembly of Gradient Carbon Dot Hydrogel Electrolytes for Stable Zinc Battery

6/10 · *Advanced Materials* · operando electrochemical–confocal fluorescence microscopy

Couples a confocal fluorescence platform with an electrochemical cell to track fluorescent carbon dots migrating electrophoretically in a hydrogel electrolyte and enriching at the Zn interface. The visualization is used to build a gradient electrolyte. The mechanism rationale still leans on FEM and DFT, and the additive is zinc-specific.

9. Electrochemically In Situ Formed Inverse-Honeycomb Local Ordering for Low-Voltage-Hysteresis Oxygen Redox in Na/Li Layered Oxide Cathodes

6/10 · *Angewandte Chemie International Edition* · atomic-resolution STEM, ex situ spectroscopy, thermal relaxation, ion-exchange

Atomic-resolution STEM shows that deep desodiation of $NaLi_{0.2}Ni_{0.2}Mn_{0.6}O_2$ forms an inverse-honeycomb superstructure in the alkali layer. Comparing Li analogues made by ion exchange from pristine and charged precursors, plus thermal relaxation and spectroscopy, ties this ordering to suppressed Ni-centred ligand-to-metal charge transfer and lower voltage hysteresis. The evidence is mostly ex situ.